

Junan Lin

PhD Student in Oxford Control Group | ✉ junan.lin@eng.ox.ac.uk

Interests: learning-to-optimize, convex optimization, ADMM/OSQP, model predictive control

Education

University of Oxford, DPhil in Engineering Science (fully funded) Oct 2025 – Apr 2029 (expected)

- Supervisors: Prof. Luca Furieri and Prof. Paul Goulart

ETH Zürich, M.Sc in Robotics, Systems and Control, GPA 5.77/6.00 (distinction) Sep 2022 – May 2025

- Relevant coursework: Model Predictive Control, Linear System Theory, Machine Perception, 3D Vision, Planning and Decision Making, Robot Dynamics

Beihang University, B.Eng in Automation, GPA 3.82/4.00 (top 10%) Sep 2018 – Jun 2022

Publications and Preprints

Junan Lin, Paul J. Goulart, and Luca Furieri, “Learning Over-Relaxation Policies for ADMM with Convergence Guarantees”, *arXiv*, 2026.

Junan Lin, *et al.*, “Enhancing Port Automation: A Novel Object Detection Pipeline for Container Ship Bays”, *IEEE SENSORS*, Kobe, Japan, 2024.

Junan Lin, *et al.*, “CaSAR: Contact-aware Skeletal Action Recognition”, *arXiv*, 2023.

Junan Lin, *et al.*, “Policy Optimization for Multi-scale Melt Pool Feedback Control in Laser Powder Bed Fusion”, manuscript in preparation.

Research Experiences

Learning Over-Relaxation Policies for ADMM Oct 2025 – Mar 2026, [📄 Paper](#)

- Designed and implemented **neural network policies** for adaptive optimization in ADMM to improve solver efficiency while preserving **convergence guarantees**.
- Improved convergence speed of OSQP by **10–20%** on benchmark QPs (e.g. Portfolio, SVM, MPC) without introducing additional refactorization.

Learning Constrained Value Functions for MPC Oct 2024 – May 2025, [📄 Slides](#)

- Developed a data-driven optimization framework for value function approximation in constrained optimal control, targeting fast online decision-making for dynamical systems.
- Computed value, gradient, and Hessian of the value function efficiently via the **envelope theorem**, avoiding costly neural network training.
- Achieved **5×** speedup compared to long-horizon MPC while maintaining high feasibility (**up to 96% vs 38% baseline**).

Policy Optimization for Melt Pool Control in Selective Laser Melting Apr 2024 – Jul 2025, [📄 Slides](#)

- Developed a control scheme that combines **temperature trajectory optimization, iterative learning control, and output feedback control (designed by policy gradient)** to drive the surface temperature of selective laser melting system to a reference temperature.
- Achieved performance comparable to Bayesian optimization **on real hardware** with **simulation-trained** parameters.

Internship Experiences

ABB Schweiz Oct 2023 – Mar 2024 | May 2025 – Aug 2025

Computer Vision & Machine Learning R&D Intern, Baden, Switzerland [📄 Paper](#)

- Designed and implemented an end-to-end **learning pipeline** for port automation, transforming 3D sensing data into structured 2D representations for object detection with **Deformable DETR**.
- Achieved **97% precision & recall** in container detection and **under 5 cm** error in 3D position estimation.
- Built simulation-based learning workflows and developed **reinforcement learning** policies with curriculum learning for large-scale container loading scenarios.

DVSense (Beijing) technology June 2023 – September 2023

Computer Vision R&D Intern, Zürich, Switzerland

- Built OCR pipelines and improved deployment efficiency by **2–4×** using ONNX/TensorRT.

Skills

Programming: Python (NumPy, SciPy, Pandas, PyTorch), C/C++, MATLAB.

Scientific computing: CVXPY, CasADi. **Tools:** Git, ONNX, TensorRT, Isaac Sim.